

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
1. Given:
Voltage (V) = 12 volts
Power (P) = 40 watts
We need to find: Current (I) and Resistance (R)
Formula : $P = V * I$
Rearrange to find I : $I = P / V$
 $I = 40 \text{ watts} / 12 \text{ volts} = 3.33 \text{ amps}$
Now that we have I , we can use the formula : $V = I * R$
Rearrange to find R : $R = V / I$
 $R = 12 \text{ volts} / 3.33 \text{ amps} = 3.6 \text{ ohms}$
So , the current drawn by the head light bulb is 3.33 amps , and its resistance must be 3.6 ohms .
</think><answer> E. 3.33 amps , 3.6 ohms